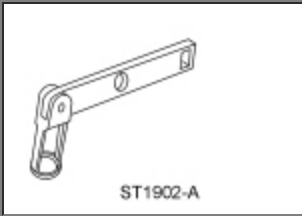
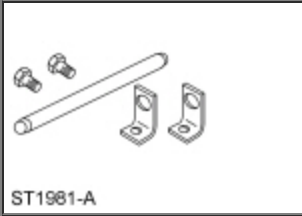
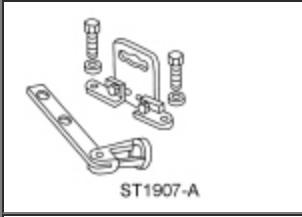
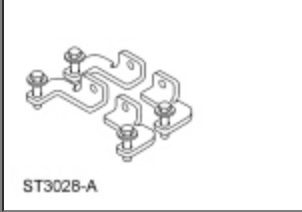


Cylinder Head

 [Ford Ecat Electronic Parts Catalog](#)

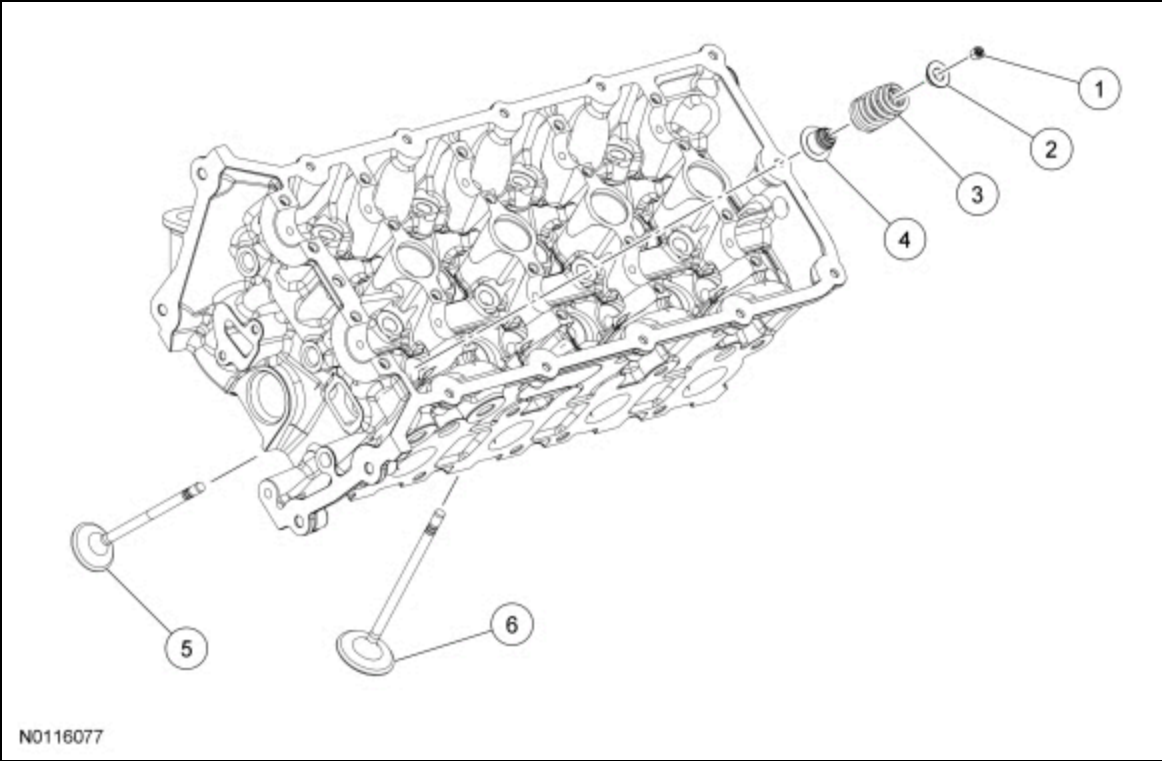
Special Tool(s)

 ST1902-A	Compressor, Valve Spring 303-1418
 ST1981-A	Compressor, Valve Spring 303-300 (T87C-6565-A)
 ST1907-A	Compressor, Valve Spring 303-350 (T89P-6565-A)
 ST3028-A	Compressor, Valve Spring 303-1249

Material

Item	Specification
Motorcraft® SAE 5W-20 Synthetic Blend Motor Oil (US); Motorcraft® SAE 5W-20 Super Premium Motor Oil (Canada) XO-5W20-QSP (US); CXO-5W20-LSP12 (Canada)	WSS-M2C945-A

NOTE: LH cylinder head shown, RH cylinder head similar.



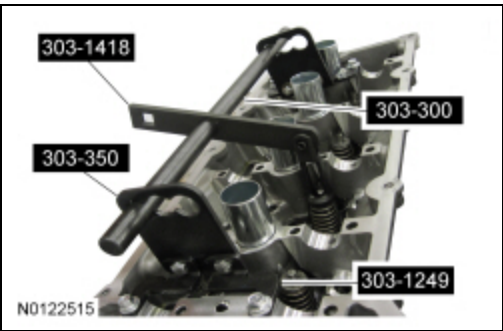
Item	Part Number	Description
1	6518 6518	Valve spring retainer key (32 required)
2	6514 6514	Valve spring retainer (16 required)
3	6513 6513	Valve spring (16 required)
4	6571 6571	Valve seal (16 required)
5	6507 6507	Intake valve (8 required)
6	6505 6505	Exhaust valve (8 required)

Disassembly

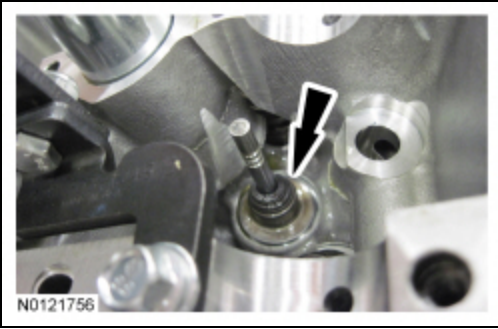
NOTICE: During engine repair procedures, cleanliness is extremely important. Any foreign material, including any material created while cleaning gasket surfaces, that enters the oil passages, coolant passages or the oil pan, can cause engine failure.

NOTE: If the components are to be reinstalled, they must be installed in their original locations. Mark the components for installation into their original locations.

1. Using the Valve Spring Compressors, remove the keys, retainer and spring.



2. Remove and discard the valve seal.



3. Remove the valve from the cylinder head.
4. Repeat the previous steps for each of the remaining valves.

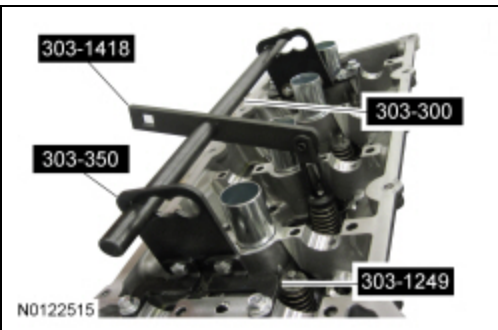
Assembly

NOTE: Apply clean engine oil to the valve seals and the inside diameter of the valve guides prior to assembling the cylinder head components.

1. **NOTE:** The valve seal must be bottomed out on the cylinder head spring seat.

Install the valve seal.

2. Install the valve into the cylinder head.
3. Using the Valve Spring Compressors, install the spring, retainer and keys.



4. Repeat the previous steps for each of the remaining valves.